AI-Driven Fabric Defect Detection & Monitoring System for the Apparel Industry in SRI LANKA

Progress Presentation 2

25-26J-452

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PROBLEM & IMPACT

! Current Industry Problems

- *Fabric Inspection is manual*
- *Operators work at high speed*
- *Defects detected late*

Result In

- *Fabric Wastage*
- *Higher Production Cost*
- *Quality Complaints & Rejections*

Sub Problem

- *Continuous fabric capturing generates huge volumes of redundant frames.*
- *Traditional systems capture every frame, Increasing latency and cost.*
- *No Intelligence at Capture Stage.*

Proposed Solutions

1. **Motion-Driven Frame Capture**
   (Frames captured only when fabric motion is detected)

2. **Unsupervised Anomaly Pre-Screening**
   (Autoencoder learns normal fabric behavior and flags only irregular frames)

3. **Frame-Level Quality & Routing**
   (Each suspicious frame is scored for quality and routed intelligently)

Sub Problem

- *Existing systems use fixed preprocessing, which cannot adapt to different fabric types or lighting conditions.*
- *Defect detection is often applied directly on unstable raw images at the edge.*
- Edge-level image quality stabilization is underexplored.

User Requirements

- Real-time, low-latency edge-level enhancement..
- Automatic adaptation to different fabric types.
- Forward only stabilized images for defect detection.

Proposed Edge-Level Solutions

- **ML-driven fabric-aware enhancement** that adapts settings in real time.
- **AI-based enhancement evaluation** using quality metrics and frame scores.
- **Operator dashboard** with visual analytics, alerts, and enhancement metrics.

Problem Definition

- Current fabric quality control systems only detect defects and provide generic alerts without exact locations.
- Operators must manually search fabric rolls, resulting in time loss and reduced productivity.
- Manual inspection is prone to human error and inconsistency, leading to missed defects and quality variations.

Proposed Edge-Level Solutions

- ML-Driven Fabric defect detection with YOLOv9, detects defects like stains and holes.
- Encoder-Based Fabric Tracking: Custom rotary encoder with measuring wheel tracks fabric movement.
- Real-Time Defect Localization: Enables accurate location of defects on fabric rolls.

User Requirements

- Automatic detection and classification of fabric defects.
- Measure fabric length with exact defect positions using rotary encoder.
- Real-time visual alerts for immediate operator action.

Sub Problem

- Detect defects only after they occur, with no prediction of upcoming defect locations.
- Slow, error-prone, reducing proactive quality control.
- No risk scoring or root cause analysis to identify machine, material, or operator issues.

Proposed Solutions

- ML-Based Defect Prediction (LSTM): upcoming defect locations in real time.
- Predictive Risk & RCA (XGBoost and Random Forest): risk scores, defect pattern.
- Operator-Centric Insights: inspection zones, urgency levels, recommended actions.

User Requirements

- Ability to Predict future defect locations in real time.
- Provide risk scores and defect/root cause classification
- Support fast, edge-ready dashboard visualization.

Commercialization

Customer Persona

- **Large & medium** garment / textile manufacturers.
- Quality managers, production supervisors, factory owners

Market Size

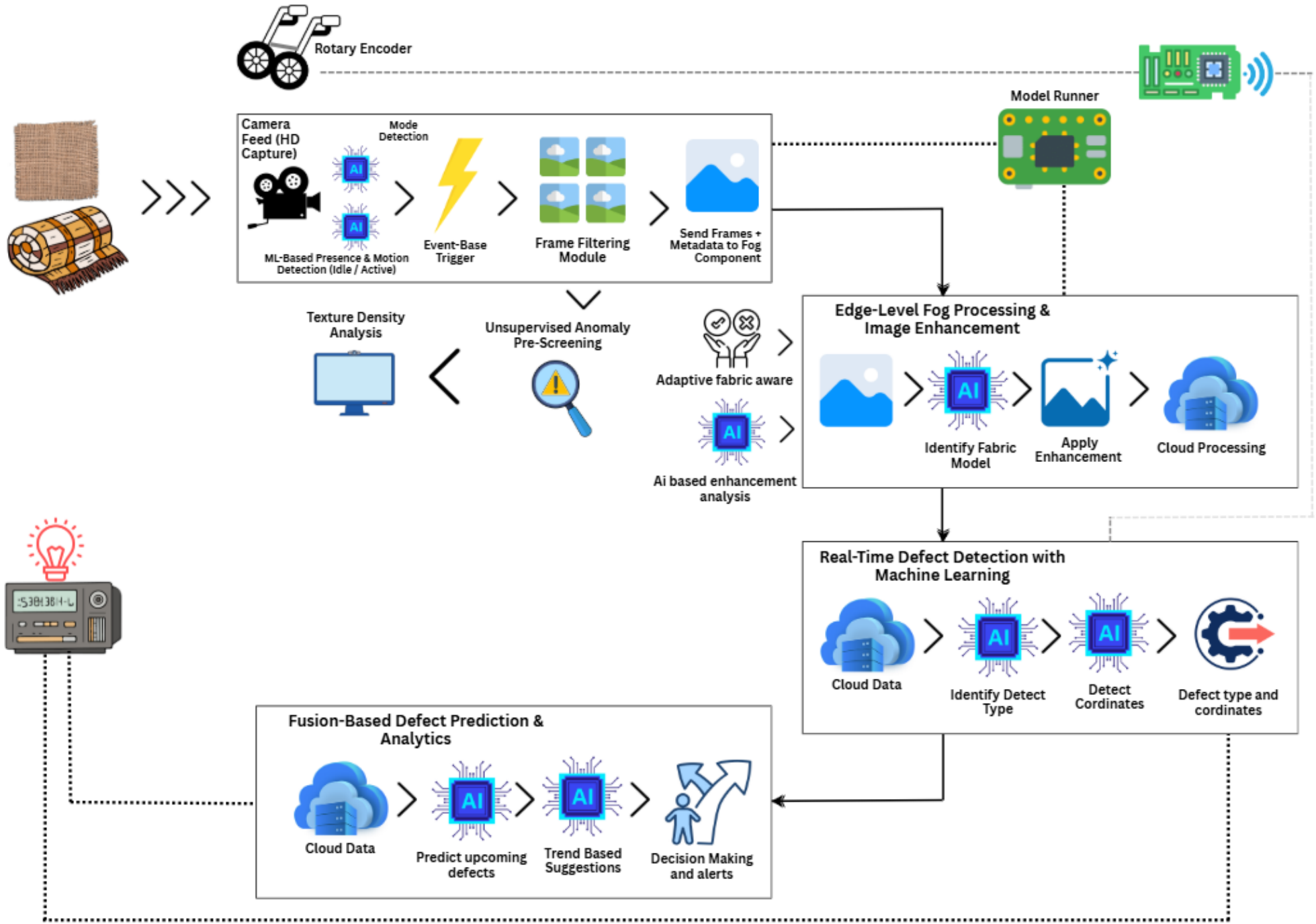
- Global textile & garment industry: USD 1+ trillion.
- Growing demand for automation, AI quality inspection

What is Unique in Our Solution

- **End-to-end pipeline**: capture → enhancement → defect detection → quality intelligence.
- **Prevents misleading defect analysis** using quality gating.

Cost Recovery / Revenue Model

- **Subscription (SaaS)** for analytics dashboard.
- Per-line / per-roll licensing.


Rotary Encoder
Model Runner
Camera Feed (HD Capture)
Mode Detection
ML-Based Presence & Motion Detection (Idle / Active)
Event-Base Trigger
Frame Filtering Module
Send Frames + Metadata to Fog Component
Texture Density Analysis
Unsupervised Anomaly Pre-Screening
Adaptive fabric aware
Ai based enhancement analysis
Edge-Level Fog Processing & Image Enhancement
Identify Fabric Model
Apply Enhancement
Cloud Processing
Real-Time Defect Detection with Machine Learning
Cloud Data
Identify Detect Type
Detect Cordinates
Defect type and cordinates
Fusion-Based Defect Prediction & Analytics
Cloud Data
Predict upcoming defects
Trend Based Suggestions
Decision Making and alerts